

Problem 14 – Smart Employee Payroll System

Difficulty: ★★☆☆ Intermediate

Overview

Build a **Streamlit Employee Payroll System** that allows HR to calculate salaries, view the most recent payroll, and review payroll records for all processed employees.

 **Streamlit only:** Do not use `input()`, `print()`, `while True`, or console menus.

1. Employee Data

Store employees in a **list of dictionaries**.

ID	Name	Hourly Rate
1001	Ahmed	120 EGP
1002	Sara	150 EGP
1003	Omar	180 EGP
1004	Mariam	200 EGP

Each employee contains:

- Employee ID
- Employee name
- Hourly rate

2. Application Layout

Sidebar

Display:

- Payroll formulas
- Company working hours
- Payroll policy

Tabs

Create three tabs:

1. **Payroll Calculator**
 2. **Recent Payroll**
 3. **All Employees Payroll**
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3. Payroll Calculator

Place all payroll inputs inside an `st.form()`.

Input	Widget	Requirement
Employee	<code>st.selectbox()</code>	Select employee
Working Hours	<code>st.number_input()</code>	> 0
Overtime Hours	<code>st.number_input()</code>	≥ 0
Bonus	<code>st.number_input()</code>	≥ 0
Deduction	<code>st.number_input()</code>	≥ 0

Use `st.form_submit_button()` as the **primary Calculate Payroll** action.

Use suitable labels, minimum/step values, and appropriate icons.

4. Validation

When submitted:

- Employee must be selected.
- Working hours must be greater than 0.
- Overtime hours must be 0 or greater.
- Bonus must be 0 or greater.
- Deduction must be 0 or greater.

Invalid information → `st.error()`.

Overtime greater than **20 hours** → `st.warning()`.

5. Salary Calculation

Basic Salary = Hourly Rate × Working Hours

OT Pay = Hourly Rate × Overtime Hours × 1.5

Gross Salary = Basic Salary + OT Pay + Bonus

Net Salary = Gross Salary – Deduction

Display all amounts in **EGP**.

6. Performance ★

Overtime Hours	Performance
0	Standard
1–10	Good
11–20	Excellent
> 20	Outstanding

Display the performance in the payroll results.

7. Process & Store Payroll

After a valid submission:

- Calculate the salary.
- Determine performance.
- Store the payroll in a **list of dictionaries**.
- Save payroll records using `st.session_state`.
- Make the latest payroll available in **Recent Payroll**.
- Use `st.spinner()` while processing.
- Use `st.success()` after completion.

Each payroll record should contain:

- Employee ID
- Employee name
- Hourly rate
- Working hours
- Overtime hours

- Basic salary
 - OT Pay
 - Bonus
 - Gross salary
 - Deduction
 - Net salary
 - Performance
-

8. Recent Payroll Tab

Display the **most recently processed payroll**.

If no payroll exists, use `st.info()`.

Use **two bordered columns**:

Left Column

Display:

- Employee ID
- Employee name
- Hourly rate
- Working hours
- Overtime hours

Right Column

Display:

- Basic Salary
- OT Pay
- Bonus
- Gross Salary
- Deduction
- Net Salary
- Performance

Required Formatting

- **OT Pay** → green
- **Bonus** → green
- **Deduction** → red
- **Net Salary** → **bold**
- **Performance** → **bold**

Use appropriate Streamlit text formatting such as `:green[...]`, `:red[...]`, and bold Markdown.

Use `st.code()` to display a formatted payroll summary.

9. All Employees Payroll Tab

Display a table containing **all processed payroll records**.

Each row represents one processed payroll.

The table must include:

- Employee ID
- Employee
- Hourly Rate
- Working Hours
- Overtime Hours
- Basic Salary
- OT Pay
- Bonus
- Gross Salary
- Deduction
- Net Salary
- Performance

The table must update whenever a new payroll is calculated.

If no payroll has been processed, display an appropriate `st.info()` message.

The table represents **processed payrolls**, not simply the original employee list.

10. Required Text Functions

Use all six:

- `st.title()` → Application title
- `st.header()` → Main sections
- `st.subheader()` → Smaller sections
- `st.write()` → General information/results
- `st.code()` → Recent payroll summary
- `st.divider()` → Separate sections

The main title must be **centered** using:

```
text_alignment="center"
```

11. Required Input Widgets

Use:

- `st.selectbox()` → Employee
- `st.number_input()` → Working hours
- `st.number_input()` → Overtime
- `st.number_input()` → Bonus
- `st.number_input()` → Deduction
- `st.form()` → Payroll form
- `st.form_submit_button()` → Calculate Payroll

The Calculate Payroll button must be the **primary** action.

Use appropriate labels, limits, steps, and icons.

12. Required Status Elements

Use all five:

- `st.info()` → Information / no payroll
 - `st.success()` → Payroll completed
 - `st.warning()` → Overtime exceeds 20 hours
 - `st.error()` → Invalid payroll information
 - `st.spinner()` → Payroll processing
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13. Competition UI Requirements

The application must demonstrate:

- Centered main title.
- Appropriate text colors using `:color-name[Text]`.
- Primary button for the main payroll action.
- Bordered columns using `st.columns(2, border=True)`.
- Appropriate input icons.
- `st.divider()` between major sections.
- Clear alignment and grouping.
- Readable presentation suitable for a large screen.

Required Result Formatting

Green:

- OT Pay
- Bonus

Red:

- Deduction

Bold:

- Net Salary
 - Performance
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14. Required Functions

Create functions for:

- **Find Employee**
- **Calculate Payroll**
- **Calculate Performance**
- **Display Recent Payroll**
- **Display All Payrolls**

You may create additional functions when necessary.

Challenge Mission

Monthly Payroll Dashboard

Using the stored payroll records, display:

- Total employees processed
 - Total salaries paid
 - Average salary
 - Highest salary
 - Lowest salary
 - Employee with highest overtime
 - Total bonuses paid
 - Total deductions
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Python Topics

- Variables
 - Strings
 - Lists
 - Dictionaries
 - Integers and floats
 - Arithmetic operators
 - Comparison operators
 - Assignment operators
 - Conditional statements
 - `for` loops
 - User-defined functions
 - List and dictionary operations
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Streamlit Topics

Displaying Text

- `st.title()`
- `st.header()`
- `st.subheader()`
- `st.write()`
- `st.code()`
- `st.divider()`

Input

- `st.selectbox()`
- `st.number_input()`
- `st.form()`
- `st.form_submit_button()`

Status

- `st.info()`
- `st.success()`
- `st.warning()`
- `st.error()`
- `st.spinner()`

Layout & State

- `st.sidebar`
- `st.tabs()`

- `st.columns()`
 - `st.session_state`
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Final Checklist

- ☐ Employee data stored as dictionaries
 - ☐ Payroll records stored as dictionaries
 - ☐ `st.session_state` used
 - ☐ Sidebar
 - ☐ Payroll Calculator tab
 - ☐ Recent Payroll tab
 - ☐ All Employees Payroll tab
 - ☐ Payroll form
 - ☐ Validation
 - ☐ Salary calculations
 - ☐ Performance classification
 - ☐ Recent payroll displayed in bordered columns
 - ☐ All processed payrolls displayed in a table
 - ☐ OT Pay green
 - ☐ Bonus green
 - ☐ Deduction red
 - ☐ Net Salary bold
 - ☐ Performance bold
 - ☐ Centered title
 - ☐ Primary Calculate button
 - ☐ Input icons
 - ☐ Dividers
 - ☐ All required status elements
 - ☐ Monthly dashboard challenge
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Important Restrictions

Use **Streamlit only** for user interaction.

Do not use:

- `input()`
- `print()`
- `while True`
- Console menus
- Console interaction